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ANTIBIOTIC TREATMENT GUIDELINES FOR DEEP NECK SPACE INFECTIONS IN PATIENTS ON PEDIATRIC SERVICES

This guideline is designed to provide guidance in pediatric patients with a primary deep neck space infection. Management of deep neck space infections in patients < 2 months of age, patients who are severely immunocompromised, or those presenting with septic shock is beyond the scope of these guidelines. For undifferentiated sepsis with organ dysfunction or septic shock, refer to the Pediatric Sepsis Guidelines.

Indication	Empiric Therapy	Duration/Comments
Bacterial lymphadenitis -- without suspected dental source Target Pathogens: *Staphylococcus aureus,* Group A *Streptococcus*	Outpatient Therapy or Step-down (from IV to PO) Therapy *1st line if no MRSA risk factors present[1]:* **Cephalexin*** 25 mg/kg/DOSE PO TID (max: 1 g/DOSE) *If MRSA risk factors present[1] and negative for Group A Streptococcus:* **TMP-SMX[2]*** 6 mg TMP/kg/DOSE PO BID (max: 320 mg TMP/DOSE) *If MRSA risk factors present[1] and positive for Group A Streptococcus:* **Amoxicillin*** 25 mg/kg/DOSE PO BID (max: 1 g/DOSE) + **TMP-SMX[2]*** 6 mg TMP/kg/DOSE PO BID (max: 320 mg TMP/DOSE) *Allergy that precludes the use of amoxicillin[7], cephalexin[7], or TMP/SMX[2]:* **Clindamycin** 10 mg/kg/DOSE PO TID (max: 450 mg/DOSE) See comment to the right regarding *S. aureus* resistance rates -OR- **Linezolid[8]*** PO: <12 years: 10 mg/kg/DOSE TID (max: 600 mg/DOSE) ≥12 years: 10 mg/kg/DOSE BID (max: 600 mg/DOSE) Inpatient (IV) Therapy: *1st line:* **Cefazolin*** 33 mg/kg/DOSE IV q8h (max: 2 g/DOSE) + **Collect MRSA nasal swab** *If MRSA nasal swab positive, MRSA risk factors present[1], or allergy that precludes cefazolin use[7]:* **Vancomycin*** IV	If lack of response to empiric therapy, please consider need for drainage, discussion with consultants (ENT), and consider other pathogens including cat scratch disease (*Bartonella henselae*) and non-tuberculous mycobacteria. Michigan Medicine pediatric *S. aureus* resistance rates are low for TMP/SMX[2] (2%) and linezolid (0%), compared to clindamycin (26% for MSSA and 40% for MRSA in 2024). Duration: -- 10 days without surgical drainage -- If surgical drainage is performed, 5 days from procedure • May extend therapy up to 7-10 days if lack of symptom resolution at 5 days
Bacterial lymphadenitis -- with suspected dental source Target Pathogens: oral anaerobes, oral streptococci	Outpatient Therapy or Step-down (from IV to PO) Therapy: *1st line:* **Amoxicillin-clavulanate[3]*** 25 mg amoxicillin/kg/DOSE PO BID (max: 875 mg amoxicillin/DOSE) *Low-risk[4] penicillin allergy:* **Cephalexin*** 25 mg/kg/DOSE PO TID (max: 1000 mg/DOSE) + **Metronidazole*** 10 mg/kg/DOSE PO TID (max: 500 mg/DOSE) *Allergy that precludes use of amoxicillin-clavulanate and cephalexin[7]:* **Clindamycin** 10 mg/kg/DOSE PO TID (max: 450 mg/DOSE) Inpatient (IV) Therapy: *1st line:* **Ampicillin-sulbactam*** 50 mg ampicillin/kg/DOSE IV q6h (max: 2 g ampicillin/DOSE) *Low-risk[4] or high-risk[5] penicillin allergy:* **Cefazolin*** IV + **Metronidazole*** 10 mg/kg/DOSE IV q8h (max: 500 mg/DOSE) *Allergy that precludes use of ampicillin-sulbactam and cefazolin[7]:* **Clindamycin** 10 mg/kg/DOSE IV q8h (max: 600 mg/DOSE)	If lack of response to empiric therapy, please consider need for drainage, discussion with consultants (ENT, OMFS). Duration: -- 10 days without surgical drainage -- If surgical drainage is performed, 5 days from procedure • May extend therapy up to 7-10 days if lack of symptom resolution at 5 days

Lymphadenitis, concern for non-tuberculous mycobacteria (NTM)	Excision is the preferred treatment for NTM lymphadenitis. Please consult ENT for management.	
Peritonsillar abscess or Retropharyngeal abscess Target Pathogens: Group A *Streptococcus, Staphylococcus aureus, Haemophilus* species (rare), oral anaerobes, including *Fusobacterium necrophorum* (more common in retropharyngeal abscess); often polymicrobial	Outpatient Therapy or Step-down (from IV to PO) Therapy: *1st line:* **Amoxicillin-clavulanate[3]*** 25 mg amoxicillin/kg/DOSE PO BID (max: 875 mg amoxicillin/DOSE) *Low-risk[4] or high-risk[5] penicillin allergy:* **Cefuroxime*** 15 mg/kg/DOSE PO BID (max: 500 mg/DOSE) + **Metronidazole*** 10 mg/kg/DOSE PO TID (max: 500 mg/DOSE) *High-risk[5] cephalosporin allergy or beta-lactams contraindicated[6]:* **Levofloxacin***: <5 years: 10 mg/kg/DOSE PO BID (max: 375 mg/DOSE) ≥5 years: 10 mg/kg/DOSE PO daily (max: 750 mg/DOSE) + **Metronidazole*** 10 mg/kg/DOSE PO TID (max: 500 mg/DOSE) *If MRSA risk factors present[1], ADD:* **TMP-SMX[2]*** 6 mg TMP/kg/DOSE PO BID (max: 320 mg TMP/DOSE) Inpatient (IV) Therapy: *1st line:* **Ampicillin-sulbactam*** 50 mg ampicillin/kg/DOSE IV q6h (max: 2 g ampicillin/DOSE) *Low-risk[4] or high-risk[5] penicillin allergy:* **Cefuroxime*** 50 mg/kg/DOSE IV q8h (max: 2 g/DOSE) + **Metronidazole*** 10 mg/kg/DOSE IV q8h (max: 500 mg/DOSE) *High-risk[5] cephalosporin allergy or beta-lactams contraindicated[6]:* **Levofloxacin***: <5 years: 10 mg/kg/DOSE IV q12h (max: 375 mg/DOSE) ≥5 years: 10 mg/kg/DOSE IV q24h (max: 750 mg/DOSE) + **Metronidazole*** 10 mg/kg/DOSE IV q8h (max: 500 mg/DOSE) *If MRSA risk factors present[1] or severe airway threatening disease, ADD:* **Vancomycin*** IV	Otolaryngology surgical consultation is recommended in patients with peritonsillar abscess and retropharyngeal abscess. If cultures are obtained, tailor antibiotic therapy to results of Gram stain, culture, and sensitivities. Cefuroxime oral suspension must be compounded (send to Taubman pharmacy); tablets may be crushed. Duration: -- 10 days from surgical drainage with clinical improvement -- 14 days if no drainage is performed
Lemierre's Syndrome Target Pathogen: *Fusobacterium necrophorum*	Lemierre's syndrome is extension of a bacterial pharyngitis/tonsillitis into the lateral neck spaces, with subsequent septic thrombophlebitis of the internal jugular vein. Please consult Pediatric Infectious Diseases to discuss empiric antibiotics if concerned for Lemierre's syndrome. For patients with septic shock or sepsis with organ dysfunction, refer to the Pediatric Sepsis Guidelines.	

Footnotes:

*Renal adjustment may be necessary. See Pediatric Antimicrobial Dosing Guidelines.

[1] Consider MRSA coverage if any of the following are present: sepsis with organ dysfunction or septic shock, immunocompromised status, personal or household contact with MRSA infection, or colonization in the past 12 months.

[2] TMP-SMX: trimethoprim-sulfamethoxazole

[3] For standard dosing, 7:1 amoxicillin-clavulanate formulations are preferred. This includes 875-125 mg tablets, 400-57 mg/5 mL oral suspension, or 200-28.5 mg/5 mL oral suspension.

[4] Low-risk allergies include: pruritus without rash, urticaria/hives with no other symptoms, remote (>10 years) unknown reaction, patient denies allergy but is on record, mild to severe rash with no other symptoms (mild rash: non-urticarial rash that resolves without medical intervention).

[5] High-risk allergies include: anaphylaxis, respiratory symptoms, angioedema/swelling, cardiovascular symptoms (syncope/passing out, arrhythmia).

[6] Severe non-IgE allergy: organ damage (kidney, liver), drug-induced immune-mediated anemia/thrombocytopenia/leukopenia, rash with mucosal lesions (Stevens Johnsons Syndrome/Toxic Epidermal Necrolysis), rash with pustules (acute generalized exanthematous pustulosis), rash with eosinophils and organ injury (DRESS – drug rash eosinophilia and systemic symptoms), rash with joint pain, fever, and myalgia (Serum Sickness).
[7] See Beta-lactam allergy evaluation and empiric guidance for further information.
[8] Serotonin Syndrome and Linezolid: Education and Recommendations

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	Citation
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Revision History

Date	Revision

The recommendations in this guide are meant to serve as treatment guidelines for use at Michigan Medicine facilities. If you are an individual experiencing a medical emergency, call 911 immediately. These guidelines should not replace a provider's professional medial advice based on clinical judgement, or be used in lieu of an Infectious Diseases consultation when necessary. As a result of ongoing research, practice guidelines may from time to time change. The authors of these guidelines have made all attempts to ensure the accuracy based on current information, however, due to ongoing research, users of these guidelines are strongly encouraged to confirm the information contained within them through an independent source.